Herd Immunity and Learning Externalities in Markets of Adaptive Models

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Abstract

A learning agent can pass every single-agent stability test and still help destabilize the ecosystem of adaptive models it runs in: when agents share an environment that reacts to their decisions, each one's retraining reshapes the data the others learn from next and no agent internalizes that effect. We call it a *learning externality*. Its strength is the market's systemic modulus m_N, the factor by which a common perturbation grows per retraining round. What sets m_N is not the number of firms but their *effective crowding*, the leading eigenvalue of the correlation matrix of firms' response Jacobians, which counts firms per independent model. Foundation-model concentration therefore enters the stability condition as a term rather than as a concern: fifty firms fine-tuning one vendor's model are dynamically close to one large learner. Three closed-form levers follow from a reduction lemma we derive in the main text. Taking fewer gradient steps between deployments buys stability, but only up to a critical crowding level past which no cadence is safe. Feedback-aware correction reproduces the imperfect-vaccine coverage law, stabilizing the market once the corrected fraction exceeds $(1 - 1/m_N)/e$ at efficacy e, a form exact in the fully shared limit and accurate away from it. A Pigouvian wedge prices the externality, and every market of two or more firms over-adapts. Correction and model diversity are substitutes along a frontier we compute. In a shared-pool order-flow market the mechanism reproduces the $1.74\times$ and $3.16\times$ amplification of a recent preprint, bit for bit.

1 Introduction

A learning agent can pass every stability test we know how to run and still help destabilize the market it operates in.

The test we know how to run is a single-agent test. A model is evaluated, stress-tested and certified alone, against a distribution treated as given. When the deployed model reshapes that distribution, the performative prediction literature supplies the correction: repeated retraining converges if and only if a modulus $m = \varepsilon\beta/\gamma$ stays below one, where ε measures how strongly the environment responds and γ is the learner's own objective curvature (Perdomo et al., 2020). Certify $m < 1$ and the loop is safe.

It is not, once the agent has company. Consider dealers whose quotes reshape a common pool of informed flow, recommender platforms fine-tuning one foundation model against one pool of user attention, or lenders pricing against one population of borrowers. Each agent's retraining changes the data every *other* agent will learn from next, and no agent internalizes that effect. We call this a *learning externality*: every participant behaves rationally, passes its own test, and helps amplify a loop none of them chose. The intuition is not new. Banks that each diversify optimally hold similar portfolios and fail together (Beale et al., 2011; Wagner, 2010), and funds running similar strategies

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unwound together in August 2007 with the crowding invisible from any single risk report (Khandani and Lo, 2007). What is new is that the agents now *learn*, which changes both what determines the danger and what can be done about it.

What determines it is not how many firms there are. Let firm i have a response Jacobian E_i describing how its own deployment reshapes the flow it faces, and let R be the correlation matrix of the vectorized E_i. We show the market's stability is governed by $m_N = N_{\text{eff}}\, m_1$ with $N_{\text{eff}} = 1 + \kappa(\lambda_{\max}(R) - 1)$, where κ is the spillover between firms and $\lambda_{\max}(R)$ is the *effective crowding*, the number of firms per independent model. With s the fraction of each response attributable to a common model, vendor or pretraining corpus, this reads $m_N = m_1(1 + \kappa s(N - 1))$, so **what enters the stability condition is the provenance of the models, not the number of firms.** The quantity a competition regulator measures is the wrong one: fifty equal-share firms have a minimal Herfindahl index and look perfectly competitive, and if all fifty share one vendor they are dynamically a monoculture (Hirschman, 1964).

What can be done about it takes three forms, and two are substitutes. Because every result is stated in m_N, each inherits the supply chain for free. *Slow down:* K-step retraining faces a window $K < K_{\max}(m_N)$ that narrows as crowding rises and shuts past a critical crowding level, beyond which no cadence is safe. *Correct:* feedback-aware updates (Izzo et al., 2021) run inside the market produce exactly the epidemiological herd-immunity law with the systemic modulus as the reproduction number. *Diversify:* that threshold rises in s, so a market reaches stability along either axis, and the iso-stability frontier between the two is this paper's most directly usable object. Pricing the externality completes the policy triple, and the decentralized equilibrium over-adapts for every $N \geq 2$. All three are ecosystem-level interventions rather than properties a single model can be certified to have, which is the limitation of the prevailing evaluation paradigm that this paper makes quantitative.

Contributions. (1) Effective crowding, firms per independent model, as a market's reproduction number, derived from a reduction lemma, which turns foundation-model concentration into a term in a stability condition and invalidates mean-similarity diversity indices twice over. (2) The crowding-cadence frontier and its critical crowding level, composing lazy deployment with multi-agent amplification. (3) A mixed-market herd-immunity theorem, exact at finite correction strength, whose clean form is the imperfect-vaccine coverage law and which carries a critical efficacy. (4) The substitution frontier between diversity and corrected learning, which exists only once (1) and (3) are expressed in one quantity. (5) The Pigouvian wedge and the over-adaptation corollary. Evidence discipline is part of the contribution rather than a caveat on it: every panel carries its status in its own row, one measured in a real order-flow market and the rest certified against the closed forms.

2 Related work

Monoculture, and why it is static. Kleinberg and Raghavan (2021) show that when many firms adopt the same algorithm welfare can fall even when the shared algorithm is the better one, because correlated selection destroys the option value of independent evaluation; Bommasani et al. (2022) formalize systemic exclusion when many deployments inherit one model. Peng and Garg (2024) is the nearest neighbour and the first to carry market effects, a two-sided matching model with one shared evaluation algorithm; Jagadeesan et al. (2023) make the supply-side move, characterizing what producers supply under a shared recommender. All four are equilibrium statements read at the market's resting point, and none expresses a stability boundary or a rate, and none carries an object for *which direction* a firm perturbs the environment, so a market can be statically fine and dynamically fragile at the same shared-model fraction.

Multiplayer performative prediction, and the containment relation. Narang et al. (2023) is the nearest dynamics: multiple learners in a shared decision-dependent environment, with equilibrium concepts and convergence conditions for repeated retraining. Their condition bounds scalars, the Lipschitz sensitivity of the joint distribution map against objective curvature, uniformly over the joint action. Ours bounds a spectral radius in which those scalars appear only inside m_1. The relation is strict refinement rather than corollary, and Proposition 3 gives the witness: two markets agreeing on every scalar of that kind, differing in stability, and so indistinguishable to any criterion built from those scalars alone. The substitution $\mathbf{1}\mathbf{1}^\top \to R$ adds an object rather than tightening a constant, and that object carries monoculture, provenance, the supply chain and every policy statement below. Two further multi-agent treatments border this one. Piliouras and Yu (2023) follow many predictors of one common outcome from global stability through instability into formal chaos as learning rates and agent counts grow, which is the monoculture corner $R = \mathbf{1}\mathbf{1}^\top$ carried past its linearization rather than one configuration among many. Li et al. (2022) schedule deployments for decentralized agents coupled by a consensus constraint, so their coupling is chosen cooperatively and no externality arises. The single-learner lazy-deployment slope of Mendler-Dünner et al. (2020) is what Theorem 4 composes with amplification no firm chooses, a composition not previously made; the corrected update of Izzo et al. (2021) is what Theorem 5 puts into the game, having only ever been run single-learner.

Systemic risk, epidemiology, and the statistic. Beale et al. (2011) asked this of banks, but their agents do not learn, so the homogeneity is portfolio overlap rather than alignment of feedback directions and there is no second instrument to trade diversity against. Diekmann et al. (1990) converts the epidemiological parallel from analogy into identity: R_0 is *defined* as the spectral radius of a next-generation operator and m_N is that of a linearized joint retraining operator, so the two thresholds are the same statement and the vaccination law transfers with its refinements. The largest eigenvalue of a correlation matrix as an effective count is meanwhile standard in random-matrix finance (Laloux et al., 1999; Plerou et al., 2002), signal processing (Roy and Vetterli, 2007) and ecology (Hill, 1973); the novelty is the matrix it is computed from, response Jacobians rather than returns, and the condition it enters. Work reading macroeconomic state off public sentiment (Nagarajan and Mittal, 2024) estimates through an exogenous channel, while the estimator of Section 9 reads an endogenous one that sharpens as the market approaches instability, so the two face opposite sample-complexity problems near the event of interest.

3 Setup: private against systemic stability

The base result. Consider N symmetric dealers quoting the same instruments and sharing one pool of informed order flow, coupled by a spillover $\kappa \in [0, 1]$: one dealer tightening its quotes raises the toxic flow routed to every dealer, in proportion κ. Each retrains periodically on the flow it has observed, and in isolation is stable exactly when its performative modulus satisfies $m_1 < 1$. The joint retraining map has Jacobian $J = -m_1[(1-\kappa)I + \kappa\mathbf{1}\mathbf{1}^\top]$, whose dominant common mode sits at $-m_1 N_{\text{eff}}$ with $N_{\text{eff}} = 1 + \kappa(N - 1)$, so the market crosses into instability at a response strength a factor N_{eff} smaller than any individual dealer's own loop tolerates. This law is inherited scaffolding (Nagarajan and Ashok, 2026), measured and reported in Section 8; everything below replaces one object in it.

Two questions, and they are not the same question. Is each agent stable alone, $m_1 < 1$? And is the collection stable when its members interact, $m_N < 1$? A shared environment separates them, because the loop closes through that environment and nothing in dealer A's validation set contains dealer B. **Agent count is not a descriptive characteristic of a market; it enters the stability condition.** Read as economics, this is a *technological* rather than pecuniary externality (Buchanan and Stubblebine, 1962): a price movement is a transfer, but reshaping the distribution a competitor learns from changes everyone's forecasting technology, a real efficiency loss. We call m_N the *feedback reproduction number*: the factor by which a common perturbation is amplified per retraining round.

Objects and standing assumptions. Firm i has a response Jacobian $E_i \in \mathbb{R}^{d\times d}$. The *alignment matrix* $R = (r_{ij})$ collects $r_{ij} = \langle \operatorname{vec} E_i, \operatorname{vec} E_j\rangle / (\|E_i\|_F \|E_j\|_F)$, a correlation matrix of feedback directions rather than of returns; it is positive semidefinite with unit diagonal.

Assumption 1 (Standing)**.** (A1) All spectral statements linearize the joint retraining map around the joint equilibrium, with the simulator as the nonlinear check. (A2) Theorem 2 is exact for equal moduli $m_i = m_1$; unequal moduli give a two-sided bound. (A3) Firms retrain on a common clock, taking a common number K of gradient steps between deployments. K is the cadence throughout: a larger K means a larger move per round, not more frequent retraining. (A4) Firms couple through one pool with a scalar spillover κ. No sign condition on R is assumed; the one place one is needed is stated where it is used, in Section 7.

The reduction. The substitution of R for $\mathbf{1}\mathbf{1}^\top$ is the whole paper, so it is derived rather than posited.

Lemma 1 (Reduction)**.** *Assume* (H1) *firm i's deviation from equilibrium is a scalar x_i along its unit response direction $\hat{E}_i = E_i/\|E_i\|_F$, so the state is $x \in \mathbb{R}^N$ rather than $\mathbb{R}^{Nd}$;* (H2) $\|E_i\|_F = \varepsilon$ *for every i, which is* (A2) *in the new object;* (H3) *firm i retrains against the component of the pool distortion along $\hat{E}_i$, feeling its own contribution in full and each competitor's with weight κ; and* (H4) *firm i best-responds to the felt distortion under a γ-strongly-convex objective of sensitivity β, so $x_i^+ = -(\beta/\gamma)\,\mathrm{felt}_i$. Then the joint retraining map is linear with Jacobian*

$$J \;=\; -m_1\big[(1-\kappa)I + \kappa R\big], \qquad m_1 = \varepsilon\beta/\gamma. \tag{1}$$

Proof sketch. The pool carries the aggregate distortion $Z = \sum_j x_j E_j$, so by (H3) firm i feels $\mathrm{felt}_i = \langle \hat{E}_i, x_i E_i\rangle + \kappa \sum_{j\neq i}\langle \hat{E}_i, x_j E_j\rangle$. By (H2), $\langle \hat{E}_i, E_j\rangle = \varepsilon\, r_{ij}$, and since $r_{ii} = 1$ the own term folds into the sum, leaving $\mathrm{felt}_i = \varepsilon[\,(1-\kappa)x_i + \kappa(Rx)_i\,]$. Applying (H4) coordinatewise gives (1), with m_1 the single-firm modulus of Perdomo et al. (2020). ☐

Alignment enters through (H3), and in the market this model comes from the hypothesis is literal rather than convenient. A dealer observes order flow only where it quotes. Flow that a competitor's repricing pushes into instruments the dealer does not trade never reaches its training set, so that competitor's perturbation is invisible to it rather than merely small, however large it is. Own-channel sensing is what makes R appear at all: two firms perturbing the environment in orthogonal directions do not read each other. (H1) is the hypothesis that limits scope; without it J carries a Kronecker structure on $\mathbb{R}^{Nd}$ whose radius is not in general $N_{\text{eff}} m_1$, the block-secular problem deferred in Section 9.

4 Result 1: effective crowding, not the count of firms

Counting firms is wrong, and the correction is where monoculture and supply-chain content enter. $\lambda_{\max}(R)$ runs from 1, all responses orthogonal, to N, monoculture, counting firms per independent model.

Theorem 2 (Alignment sets the reproduction number)**.** *Under Assumption 1 with equal moduli and heterogeneous response directions, the joint Jacobian is* (1) *and the market's spectral radius is*

$$m_N \;=\; N_{\text{eff}}\, m_1, \qquad N_{\text{eff}} \;=\; 1 + \kappa\big(\lambda_{\max}(R) - 1\big), \tag{2}$$

so the market is stable if and only if $m_N < 1$.

Setting $R = \mathbf{1}\mathbf{1}^\top$ recovers the symmetric multi-dealer law of Nagarajan and Ashok (2026) exactly, so the earlier result is not generalized away but located as the monoculture corner, where every firm's feedback points the same way. Four anchors follow, each checked independently and derived in Appendices B and C. Monoculture gives $N_{\text{eff}} = 1 + \kappa(N-1)$, the inherited law. Orthogonal responses, $R = I$, give $N_{\text{eff}} = 1$: a hundred firms perturbing the environment in a hundred orthogonal directions are dynamically one firm, and the simplex $r_{ij} = -1/(N-1)$ gives $1 + \kappa/(N-1)$. A supply chain of share s gives $N_{\text{eff}} = 1 + \kappa s(N-1)$, set by the *provenance* of the models rather than by the count of firms. $N_{\text{eff}} \geq 1$ always, so interaction never stabilizes a market below what its members achieve alone.

Proposition 3 (Strict refinement, with a witness)**.** *Fix $N \geq 2$, $\kappa \in (0,1]$ and m_1 with $1/(1+\kappa(N-1)) < m_1 < 1$. Compare the market at $R = I$ with the market at $R = \mathbf{1}\mathbf{1}^\top$. The two agree on every scalar a uniform Lipschitz-and-curvature criterion can read: the same ε by* (H2)*, the same γ and β by* (H4)*, hence the same m_1 and the same κ. Their radii are $m_N = m_1$ and $m_N = m_1(1+\kappa(N-1))$, which the choice of m_1 places on opposite sides of one. Any criterion measurable from those scalars alone therefore returns the same verdict for both, and is wrong about one of them.*

At $N = 20$, $\kappa = 0.8$ and $m_1 = 0.15$ the orthogonal market runs $m_N = 0.15$ and the monoculture $m_N = 2.43$: one converges, the other diverges, and no Lipschitz constant in either model distinguishes them. The object doing the separating is R, which no scalar criterion carries, so Theorem 2 refines a uniform condition rather than following from one.

The supply chain. Decompose $E_i = \sqrt{s}\, E_{\text{shared}} + \sqrt{1-s}\,\Xi_i$, where $s \in [0,1]$ is the fraction attributable to a shared foundation model, vendor or pretraining corpus. Error correlation across more than 350 hosted models is substantial and concentrated within shared providers (Kim et al., 2025), so the premise of this decomposition is observed on deployed systems rather than assumed. Alignments concentrate at $r_{ij} \to s$ with $O(1/d)$ fluctuation and an explicit failure probability, so $N_{\text{eff}} \to 1 + \kappa s(N-1)$ and every result below inherits s for free: fifty dealers fine-tuning one vendor's model amplify like $\sim 1 + 49\kappa$ stacked learners at $s \approx 1$ and barely more than one at $s \approx 0$.

Why the naive diversity index fails, twice. Mean alignment understates *clustered* alignment. Three tightly aligned firms among ten otherwise-orthogonal ones give $N_{\text{eff}} = 2.60$ at $\kappa = 0.8$ against 1.48 for the uniform configuration with the identical mean, so at $m_1 = 0.5$ the market runs $m_N = 1.30$ and is unstable while the index reports 0.74 and calls it safe with margin. The simplex is sharper still, minimizing mean alignment while carrying $\lambda_{\max} = N/(N-1) > 1$, so the mean fails as a *ranking* and not only as a level. And the failure is signed: $N_{\text{eff}} \geq 1 + \kappa(N-1)\bar{r}$ for every R, so the index never over-states risk (Appendix B).

5 Result 2: the crowding-cadence frontier

The first lever is quantity regulation, and it is the one a firm can pull alone: move less per round. A learner taking only K gradient steps per deployment realizes outer-map slope $\mu(K) = -m + c^K(1+m)$, where $c = 1 - \eta\gamma$ is the inner per-step contraction (Mendler-Dünner et al., 2020). Because c is built from the learner's own curvature and step size it does not depend on how many competitors it has, and that independence is what makes the composition below work; we verify it by measuring the realized contraction inside the joint market.

Theorem 4 (Systemic cadence window)**.** *Under Assumption 1, in the N-firm market with synchronous K-step retraining the joint deployment map is $c^K I + (1-c^K)J$, so the binding mode has slope $\mu_N(K) = -m_N + c^K(1+m_N)$ and the market is stable if and only if*

$$K \;<\; K_{\max} \;=\; \frac{\ln\big((m_N-1)/(m_N+1)\big)}{\ln c}, \tag{3}$$

with $K_{\max}$ infinite when $m_N \leq 1$ and strictly decreasing in m_N.

Every mode's slope sits at or below $c^K < 1$, so the upper side of the constraint never binds at any cadence and the binding mode is at $\lambda_{\max}$ whatever the sign pattern of R (Appendix D).

Critical crowding. The same inequality reads $m_N < (1+c^K)/(1-c^K)$, and since K is a positive integer the most permissive case is $K = 1$. A stabilizing cadence exists at all if and only if $m_N < (1+c)/(1-c)$, a factor of 9 at $c = 0.8$. Past that the market diverges at every cadence, however small the step budget. Moving less per round is the only unilateral defence, and past the critical level it runs out.

What the supply chain does to the window. Substituting Result 1 makes $K_{\max}$ a function of vendor concentration at fixed firm count: at $m_1 = 0.15$, $\kappa = 0.8$, $N = 30$ and $c = 0.8$, the window runs 20.68 steps at $s = 0.25$, 5.28 at $s = 0.5$ and 2.53 at $s = 1$. Raising vendor concentration cuts every incumbent's step budget eightfold while its competitors do nothing at all. The externality in its sharpest form is not that *a competitor's entry consumes your step budget*, but that *a competitor's choice of vendor does too*, without their entering.

What the lever costs. Cadence buys stability with staleness, and the exchange rate is the same quantity on both sides: after a retraining round the deployed model's gap to its own best response is exactly c^K times what it was, so the c^K that widens the stability margin is the staleness. A firm can price that trade from its own step size and curvature plus a supervisory estimate of m_N, which makes $K_{\max}$ the one number here a single participant can act on without coordinating with anyone.

6 Result 3: herd immunity, and diversity as its substitute

The second lever is a technology mandate, and unlike cadence it is not something a firm can pull alone to any effect. A *corrected* firm runs the feedback-aware update (Izzo et al., 2021; Miller et al.,

Figure 1: Left: the (N, s) phase diagram over 48 cells (Theorem 2). Right: the cadence window over 175 cells of the (N_{eff}, K) grid against predicted $K_{\max}$, with s contours and $c = 0.8$ (Theorem 4); the realized window is $\lfloor K_{\max} \rfloor$ and the plotted frontier continuous. Both are dry runs (Section 8).

2021), so its retraining is governed by the objective curvature γ_{PO} in place of the cobweb and its modulus is scaled by $\theta = \gamma/\gamma_{\text{PO}} < 1$. What is new is running it inside the game: a mixed market of N_b blind firms and $N - N_b$ corrected ones, corrected fraction $\rho_c = 1 - N_b/N$. Correction changes a firm's gain and not its response direction, so R is untouched.

Theorem 5 (Mixed-market stability)**.** *Under Assumption 1, with supply-chain alignment and two correction levels, the joint Jacobian's symmetric congruence is a diagonal matrix plus a positive rank-one term, so its spectrum solves a secular equation. Two distinct moduli collapse it to a quadratic, and for* $1 \leq N_b \leq N - 1$ *the radius is its larger root,*

$$\rho(J) = \tfrac{1}{2}\big(P + \sqrt{P^2 - 4Q}\big), \quad P = (1-\tilde{\kappa})m_1(1+\theta) + \tilde{\kappa} m_1(N_b + \theta N_c), \quad Q = \theta m_1^2 (1-\tilde{\kappa}) N_{\text{eff}}, \tag{4}$$

with $\tilde{\kappa} = \kappa s$ *and* $N_c = N - N_b$*. An empty block needs the single-block form instead, the quadratic otherwise leaving a phantom root that can exceed the true radius.*

N_{eff} appearing inside Q is why every statement below carries s: the mixed market inherits Result 1's crowding through the product term, not by stipulation.

The strong-correction limit, and it errs in the unsafe direction. As $\gamma_{\text{PO}} \to \infty$ the constant term vanishes and the unstable cycle runs through blind firms only, giving $N_b < N_c(s) = 1 + (1/m_1 - 1)/(\kappa s)$. That criterion taken alone is **optimistic rather than conservative**. The radius is nondecreasing in θ by Perron-Frobenius on an entrywise-nonnegative matrix, so the limit under-states it: on random draws it calls 11.8% of configurations stable that are unstable at every finite correction strength (2157 of 18313, exact 95% interval $[11.3, 12.3]\%$, sampling measure in Appendix K). The exact root is therefore the criterion and the limit its corollary, quoted with its error direction. The same monotonicity carries the reassuring half: **correction never backfires**, so no configuration exists in which un-blinding a firm destabilizes the market.

The collapse, and its honest generalization. At $\kappa = s = 1$ the limit reads $\rho_c^* = 1 - 1/m_N$, *exactly the epidemiological herd-immunity threshold* $1 - 1/R_0$ with the systemic modulus as the reproduction number. A market of 10 firms at $m_N = 2.5$ needs 60% of them un-blinded *if correction is perfect*. It is not, and at that same corner the quadratic degenerates and the stability condition solves exactly for

$$\rho_c^*(e) \;=\; \frac{1 - 1/m_N}{e}, \qquad e \;=\; 1 - \gamma/\gamma_{\text{PO}}, \tag{5}$$

with e the *correction efficacy*. That is the standard coverage requirement for an *imperfect vaccine*, of which the clean law is the perfect-efficacy corner. The clean form assumes homogeneous mixing (Fine et al., 2011), and the collapse here assumes the $\kappa = s = 1$ corner for precisely that reason.

Corollary 6 (Critical efficacy). $\rho_c^*(e)$ *exceeds one, so no corrected fraction works at all, exactly when* $\theta > 1/m_N$*. Correction is a usable lever only if* $\gamma_{\text{PO}}/\gamma > m_N$.

At $m_N = 2.5$ correction must deliver a $2.5\times$ curvature improvement or no fraction stabilizes the market. This parallels Section 5's critical crowding: each lever has a regime past which it stops working, and knowing both is what makes trading them off honest.

Figure 2: Left: stability against corrected fraction at $N = 20$, $m_1 = 0.15$, $\kappa = 0.8$, $s = 1$, across efficacies; the gap between the exact threshold and the perfect-correction limit is a result, not an error bar. Right: the substitution frontier, the (ρ_c, s) iso-stability curve, matching at 18 of 18 points. Both are dry runs (Section 8).

The synthesis. ρ_c^* increases in s wherever it is positive, so a market reaches stability along either axis. At $N = 20$, $m_1 = 0.15$ and $\kappa = 0.8$ a monoculture needs $\rho_c^* = 0.596$, which is 12 corrected firms; at $s = 0.5$ it needs five; at $s = 0.2$ it needs none at all.

The economic reading. Correction is a public good: an un-blinded firm captures a private benefit while the stability it contributes accrues to everyone, so the market free-rides below the threshold (Bergstrom et al., 1986). Model diversity has the same structure, a firm choosing an idiosyncratic vendor paying privately and supplying systemic stability for free. *The substitution is therefore between two goods that are both under-supplied*, not between a good and a bad, and the wedge of Section 7 is the instrument that prices exactly this under-supply.

7 Result 4: the Pigouvian wedge

The third lever is the price instrument. Inside the stable region the binding mode is an AR(1) with coefficient $-m_N$ driven by noise of variance σ^2, so its stationary variance is $V(m_N) = \sigma^2/(1-m_N^2)$. This is the smooth proxy for divergence risk, and it diverges exactly at the boundary the rest of the paper is about; nothing below uses more than V being positive, increasing and convex. The sign it discards is not lost: lag-1 autocorrelation is $-m_N$, so a crowded market's common mode oscillates rather than persists, which is what the supervision estimator of Section 9 reads.

Each firm picks an adaptation aggressiveness a_i, meaning cadence, learning rate or responsiveness, buying concave benefit $B(a_i)$ through $m_i = \mu(a_i)$. Firm i bears weight w_i on common-mode variance and the planner W, with $W > \sum_i w_i$ because every trade has a client bearing execution-quality variance that no dealer's objective contains. That is an assumption, and the results turn on its *sign* and never its size.

Lemma 7 (Marginal crowding share)**.** *At the symmetric point, with $\kappa > 0$ and $\lambda_{\max}(R)$ simple, $\partial m_N/\partial m_i = N_{\text{eff}} v_i^2$ where v is the leading eigenvector of the coupling matrix. The shares sum to N_{eff} rather than to one, and on exchangeable R each equals N_{eff}/N.*

Where a sign condition is needed, and where it is not. Theorem 2's radius needs none. R is positive semidefinite with unit diagonal, so $1 + \kappa(\lambda_i - 1)$ increases in λ_i and the radius sits at $\lambda_{\max}$ whatever the off-diagonal signs. What needs a sign is the *reading* of Lemma 7, that v is a vector of market shares. (A5) supplies it: entrywise nonnegative R puts the leading eigenvector in the nonnegative orthant by Perron-Frobenius, and shared vendors produce positive alignment.

Both readings matter. The *sum* is N_{eff}, so the market's total sensitivity to a uniform increase in aggressiveness is amplified by exactly the effective learner count of Result 1; the *individual* share $N_{\text{eff}} v_i^2$ collapses to N_{eff}/N on exchangeable R, against a naive commons guess of $1/N$ that is wrong by the factor N_{eff}, which at $N = 20$, $\kappa = 0.8$, $s = 1$ is 16.2.

Theorem 8 (The wedge)**.** *Under Assumption 1 and the welfare assumptions above, at a symmetric profile the private and social first-order conditions are the same equation with a different price on*

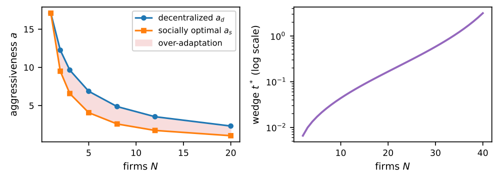

Figure 3: Over-adaptation and the wedge. Left: decentralized against socially optimal aggressiveness across N at $\chi = 1$, $s = 1$. Right: t^* against N, log scale. Dry run; the boundary divergence fits $(1 - m_N)^{-2}$ at log-log slope -2.000.

crowding, and the marginal cost a firm ignores is

$$t^* \;=\; (W - w_i)\, V'(m_N)\, \frac{N_{\text{eff}}}{N}\, \mu'(a), \qquad V'(m_N) = \frac{2\sigma^2 m_N}{(1 - m_N^2)^2}. \tag{6}$$

A per-unit fee t^* on aggressiveness makes the private condition coincide with the social one, so the taxed decentralized equilibrium implements the symmetric social optimum.

Theorem 8 characterizes the exchangeable symmetric point, which covers the monoculture and supply-chain configurations at any $s > 0$ but switches off the alignment heterogeneity Result 1 exists to represent. It excludes the orthogonal corner, where $\lambda_{\max}(R)$ is degenerate, firms decouple and there is no common mode to price. Away from it a firm sitting on the crowded mode owes more than its headcount share, and the asymmetric wedge is open (Appendix J). With symmetric weights the private condition ignores the fraction $(N-1)/N$ of the marginal variance cost borne by other firms, plus the client-side exposure entirely. t^* is strictly increasing in N, κ, s and m_N, and diverges at the boundary like $(1 - m_N)^{-2}$. The rate is the economically loaded part: a regulator setting a fee from a comfortable-looking market and holding it fixed is setting it far too low by the time the market is close to instability.

The provenance channel. m_N depends on s, so the wedge has a second channel: $\partial m_N / \partial s = m_1 \kappa (N-1)$ against $\partial m_N / \partial a_i = (N_{\text{eff}}/N)\mu'(a_i)$. The same wedge therefore prices *shared-model adoption* directly, which turns the monoculture externality into a fee rather than a warning, and the two channels are not symmetric. A firm's footprint through its own aggressiveness is bounded, since N_{eff}/N falls to κ from above, while the provenance channel is *linear in N and does not decay* against a private gain that does not grow with market size. Unregulated adoption therefore runs toward the unstable configuration, which is why the diversity floor of Section 6 is a different instrument from the cadence cap of Section 5 rather than a restatement of it (Appendix F).

Corollary 9 (Over-adaptation)**.** *The decentralized equilibrium over-adapts relative to the social optimum for every $N \geq 2$, strictly so even with client exposure switched off, and the distortion grows as the market crowds. At $N = 1$ with no client exposure the wedge is exactly zero.*

The fraction the private condition ignores is free of N_{eff}, exactly $(N-1)/N$ with no client exposure whatever κ and s are, so crowding enters the wedge's *level* through $V'(m_N)$ and not the distortion's *shape*: Corollary 9 is the generic commons result and the provenance asymmetry above is the market-of-models one. Sections 5, 6 and 7 are respectively quantity regulation, a technology mandate or structural remedy, and the price instrument; choosing among them under uncertainty is Weitzman's question (Weitzman, 1974), and we supply the objects needed to pose it rather than answering it.

8 Experiments

Six panels, each with its evidential status in its own row of Table 1, because the six are not the same kind of evidence. A panel marked *measured* ran in the order-flow simulator, a genuine N-dealer market with informed flow, spread and inventory that reduces bit for bit to the single-dealer market at $N = 1$. A panel marked *dry run* ran in a linearized reference environment realizing the model's response geometry and nothing else: it checks that the closed forms were derived from the map we believe they were derived from, which is internal validity, and it has caught real errors. What it cannot supply is external validity, evidence that a market behaves like the map. Every closed form carries a numerical certificate and the harness exits nonzero on any disagreement; Appendix I gives the

Table 1: Panel coverage and status. Panel 1 is measured in the order-flow simulator; the rest run in the linearized reference environment.

Panel	Tests	Status	Outcome
1. Amplification	Section 3	**measured**	$1.7428\times$, $3.1567\times$; relative error 0.00
2. (N, s) phase diagram	Thm 2	dry run	48 cells, max error 7.1×10^{-15}
2b. Clustered companion	Thm 2	dry run	measured m_N 1.30; mean index says 0.74
3. Cadence frontier	Thm 4	dry run	175 cells, zero disagreements
4. Herd immunity	Thm 5	dry run	thresholds $12/14/16/20$ firms across efficacy
5. Substitution frontier	Thm 5	dry run	18 of 18 points exact
6. Over-adaptation	Thm 8	dry run	12 configurations, zero contradicting Cor. 9

protocol, deterministic from a (config, seed) pair and estimating stability from realized trajectories rather than from an eigensolve. The simulator is described generically pending an artifact release, with code offered to reviewers as anonymized supplementary material.

The one measured panel. Panel 1 reproduces a recent preprint's run in that preprint's own simulator, at the relative error Table 1 records. Its claim is the monoculture corner $R = \mathbf{1}\mathbf{1}^\top$, so it is evidence for the scaffolding the four results stand on rather than for any of the four. *The gap to the linear prediction is content, not error*: the market runs -12.9% at $N = 2$ and $+5.2\%$ at $N = 3$, with *opposite signs*, which is what a nonlinear market with saturating flow does around a linearization, and the reference environment reproduces the prediction exactly because it omits that nonlinearity.

Why the rest are not measured. A heterogeneous-response port of the order-flow simulator was built and is exact at flat exposure profiles, but its acceptance gate failed: liquidity and price impact stay shared whatever the response profiles are, leaving a residual larger than several effects these panels resolve, so we drew no figure from it and moved no status. Panel 6 is a dry run for a different reason: the simulator carries no aggressiveness choice variable and no welfare object, so no measured counterpart exists to be built. Its over-adaptation gap widens from 29.0% at $N = 2$ to 119.9% at $N = 20$ with the degenerate $N = 1$ case at exactly zero, read from realized dynamics rather than the closed form (Appendix I).

9 Limitations and conclusion

Supervision from public prices. Every quantity above is hidden from the regulator these results address, and near the boundary the system tells on itself. As m_N approaches one, the leading principal component of public quotes has lag-1 autocorrelation approaching one in magnitude and variance share growing like $1/(1-m_N)$, so a supervisor can estimate distance to instability from public prices alone and the estimate sharpens as the market nears the boundary it watches for. Consistency, sample complexity and a real-data panel are deferred, the last scoped in Appendix G as consistency evidence with a placebo rather than identification.

Limitations. The analysis is linearized around the joint equilibrium (A1), and panel 1 shows what that costs in a real market. **The heterogeneous sweep is closed-form agreement, not measurement**, which is the largest gap in the work; routing liquidity and impact through the response profiles is the identified repair. Theorem 2 is exact for equal moduli (A2) and a two-sided bound otherwise, deployment is synchronous (A3), and the $1 - 1/R_0$ collapse assumes the $\kappa = s = 1$ corner. Pricing instability through stationary variance is itself a modelling choice. Share-weighted alignment, the block-secular reduction when (H1) fails, the sharp concentration rate, heterogeneous clocks, three or more correction levels and correction that moves a response direction rather than a gain are named and not attempted, each stated in Appendix J.

Conclusion. Individual model evaluation cannot certify a market of models. The quantity separating the two questions is effective crowding, which no test of one model can see. Regulation of shared general-purpose models ties its thresholds to model scale rather than to any measurable systemic quantity; this paper supplies an ecosystem-level one, with two substitutable instruments and a price.

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A Notation and standing assumptions

Symbols marked *new* are this paper's contribution surface. Everything else is inherited from the base performative-retraining model and cited rather than re-derived.

Symbol	Meaning	Range
N	firms in the market	integer ≥ 1
d	dimension of each firm's decision	integer ≥ 1
ε	performative response strength of the environment	> 0
β	sensitivity of the objective to the induced shift	> 0
γ	own-objective curvature	> 0
m_1	single-firm retraining modulus, $\varepsilon\beta/\gamma$	stable iff < 1
κ	spillover between firms through the shared pool	$[0, 1]$
E_i	firm i's response Jacobian *(new)*	$d \times d$
r_{ij}	pairwise alignment of feedback directions *(new)*	$[-1, 1]$
R	alignment matrix (r_{ij}) *(new)*	PSD, unit diagonal
$\lambda_{\max}(R)$	firms per independent model *(new)*	$[1, N]$
N_{eff}	effective crowding, $1 + \kappa(\lambda_{\max}(R) - 1)$ *(new)*	$[1, 1 + \kappa(N - 1)]$
m_N	systemic modulus, $N_{\text{eff}} m_1$ *(new)*	stable iff < 1
s	shared-model fraction *(new)*	$[0, 1]$
K	gradient steps per deployment	integer ≥ 1
η	inner learning rate	> 0
c	inner per-step contraction, $1 - \eta\gamma$	$(0, 1)$
γ_{PO}	curvature governing the corrected dynamics	$\geq \gamma$
θ	γ/γ_{PO}, the correction factor *(new)*	$(0, 1]$
e	correction efficacy, $1 - \theta$ *(new)*	$[0, 1)$
N_b	blind (uncorrected) firms *(new)*	$0 \ldots N$
ρ_c^*	herd-immunity threshold *(new)*	$[0, 1)$
a_i	firm i's adaptation aggressiveness *(new)*	> 0
t^*	the Pigouvian wedge *(new)*	

The standing assumptions of Assumption 1 are (A1) linearization of the joint retraining map around the joint equilibrium, with the simulator as the nonlinear check; (A2) equal moduli $m_i = m_1$, under which Theorem 2 is an identity and outside which it is the two-sided bound of Proposition B.6 below; (A3) synchronous retraining on a common clock at a common cadence K; and (A4) coupling through one pool with a scalar spillover κ. The sign condition (A5), that R is entrywise nonnegative, is not a standing assumption: Theorem 2's radius does not need it, and it is invoked only in Section 7, where the reading of Lemma 7 depends on it. This appendix makes both halves of that statement precise.

B The reduction and the alignment spectrum

This appendix proves Lemma 1, Theorem 2 and Proposition 3, establishes the properties of R they use, gives the three anchors, and proves that mean-based diversity indices err with a sign.

B.1 Hypotheses and the reduction

Firm i deploys $h_i \in \mathbb{R}^d$, and its deployment reshapes the flow it faces through the response Jacobian $E_i \in \mathbb{R}^{d\times d}$. Write vec E_i for the d^2-vector form, $\|\cdot\|_F$ for the Frobenius norm, and $\hat{E}_i = E_i/\|E_i\|_F$ for firm i's unit response direction. The reduction uses four hypotheses.

(H1) Rank-one deviation. Firm i's deviation from the joint equilibrium is a scalar x_i along its own response direction, so the market state is $x \in \mathbb{R}^N$ rather than $\mathbb{R}^{Nd}$.

(H2) Equal response magnitude. $\|E_i\|_F = \varepsilon$ for every i. This is (A2) restated in the new object: the inherited response strength ε *is* the Frobenius norm of the response Jacobian.

(H3) Own-channel sensing. Firm i retrains against the component of the pool distortion lying along its own response direction, feeling its own contribution in full and each competitor's with weight κ.

(H4) Quadratic retraining. Firm i minimizes a γ-strongly-convex own objective with sensitivity β to the felt distortion, giving $x_i^+ = -(\beta/\gamma)\,\mathrm{felt}_i$.

Proof of Lemma 1. The pool carries the aggregate distortion $Z = \sum_j x_j E_j$. By (H3) the distortion firm i feels is

$$\text{felt}_i \;=\; \langle \hat{E}_i,\, x_i E_i \rangle \;+\; \kappa \sum_{j \neq i} \langle \hat{E}_i,\, x_j E_j \rangle \;=\; \varepsilon \Big[\, x_i \;+\; \kappa \sum_{j \neq i} r_{ij} x_j \Big],$$

using $\langle \hat{E}_i, E_j \rangle = \varepsilon\, r_{ij}$, which is (H2). Since $r_{ii} = 1$ the own term folds into the sum,

$$\text{felt}_i \;=\; \varepsilon \Big[\, (1-\kappa)\, x_i \;+\; \kappa \sum_j r_{ij} x_j \Big].$$

By (H4), $x^+ = -(\beta/\gamma)\varepsilon\big[(1-\kappa)I + \kappa R\big]x$, and $m_1 = \varepsilon\beta/\gamma$ is the single-firm modulus, which is (1). $\square$

The alignment enters because firm i senses the pool through its own response channel, so two firms whose deployments perturb the environment in orthogonal directions do not read each other's perturbations at all, however large those perturbations are. Hypothesis (H3) is what carries that mechanism. If every E_i is equal then $r_{ij} = 1$ throughout, $R = \mathbf{1}\mathbf{1}^\top$, and the reduction returns the inherited shared-pool law exactly, so the base model is the corner of the new one in which every firm's feedback points the same way. Certificate P1 builds random E_i, forms the retraining map from the felt distortion without ever constructing R, differentiates it numerically and compares against (1), agreeing to 5.6×10^{-17} at (N, d, κ) equal to $(3, 4, 0.8)$, $(5, 6, 0.35)$, $(8, 5, 1.0)$ and $(6, 7, 0.0)$, so both spillover endpoints are covered.

Hypothesis (H1) is the substantive one. Without it the state is $(h_1, \ldots, h_N) \in \mathbb{R}^{Nd}$ and the Jacobian carries a Kronecker structure with block (i, j) equal to $-(\beta/\gamma)c_{ij}E_j$, where $c_{ij} = 1$ for $i = j$ and κ otherwise, whose spectral radius is not in general $m_1 N_{\text{eff}}$. Reducing that object is the fully heterogeneous block-secular problem, and it is deferred.

B.2 What R is

Lemma B.1. Let V be the $N \times d^2$ matrix whose rows are the unit vectors $\operatorname{vec} \hat{E}_i$. Then $R = VV^\top$ and: (1) R is positive semidefinite, so every $\lambda_i(R) \geq 0$; (2) $r_{ii} = 1$, hence $\operatorname{tr} R = N$ and $\sum_i \lambda_i(R) = N$; (3) $|r_{ij}| \leq 1$; (4) $1 \leq \lambda_{\max}(R) \leq N$; (5) $\lambda_{\max}(R) = 1$ if and only if $R = I$; (6) $\lambda_{\max}(R) = N$ if and only if $R = D\mathbf{1}\mathbf{1}^\top D$ for a sign matrix $D = \operatorname{diag}(\pm 1)$, that is, if and only if every response is parallel or antiparallel to one common direction.

Proof. (1) and (2) are immediate from $R = VV^\top$ with unit rows. (3) is Cauchy-Schwarz. For (4), the eigenvalues are nonnegative and sum to N, so the largest is at least the mean 1 and at most the total N. For (5), $\lambda_{\max} = 1$ with nonnegative eigenvalues summing to N forces all N eigenvalues to equal 1. For (6), $\lambda_{\max} = N$ forces every other eigenvalue to zero, so R has rank one, $R = vv^\top$, and unit diagonal gives $v_i^2 = 1$. $\square$

Item (4) is the one the paper leans on: $\lambda_{\max}(R)$ is a genuine effective count, bounded below by one firm and above by N, with both ends attained.

B.3 The spectrum

Proof of Theorem 2. Write $J = -m_1 B$ with $B = (1-\kappa)I + \kappa R$. B is symmetric and shares its eigenvectors with R, with eigenvalues $\nu_i = (1-\kappa) + \kappa\lambda_i(R)$, so $\rho(J) = m_1 \max_i |\nu_i|$. By Lemma B.1(1) every $\lambda_i(R) \geq 0$, and $\kappa \leq 1$ gives $1 - \kappa \geq 0$, so every $\nu_i \geq 1 - \kappa \geq 0$. The absolute value is therefore inert and the maximum sits at $\lambda_{\max}$ rather than at $\lambda_{\min}$:

$$\rho(J) \;=\; m_1\big[(1-\kappa) + \kappa\lambda_{\max}(R)\big] \;=\; m_1\big[1 + \kappa(\lambda_{\max}(R) - 1)\big] \;=\; m_1 N_{\text{eff}}.$$

Stability of the linearized joint map is $\rho(J) < 1$; substituting $m_1 = \varepsilon\beta/\gamma$ and solving for ε gives the stated form. $\square$

Certificate P3 checks this on 384 random configurations with maximum deviation 4.7×10^{-15}. Dropping the absolute value is legitimate only because R is a Gram matrix. A symmetric unit-diagonal matrix with entries outside $[-1, 1]$ can put the radius on $\lambda_{\min}$ instead: at $N = 3$ with every off-diagonal equal to -3 the spectrum is $\{-5, 4, 4\}$ and the negative mode dominates. Cauchy-Schwarz is what rules this out for a real alignment matrix, and P3 certifies both directions.

Corollary B.2. $N_{\text{eff}} \in [1,\ 1+\kappa(N-1)]$, attained at $R = I$ and $R = \mathbf{1}\mathbf{1}^\top$ respectively.

Corollary B.3. $\rho(J) \geq m_1$, with equality if and only if $R = I$. A market of adaptive models sharing an environment is never more stable than one of its members in isolation, and exact neutrality requires fully orthogonal responses.

Corollary B.3 says the externality has a sign: coupling can only amplify feedback, never damp it, so there is no configuration in which crowding does a market a favour. Individual modes can be more damped than m_1, and the differential mode at $m_1(1-\kappa)$ is one, but the radius governs and the radius cannot fall below m_1.

B.4 Which mode binds

Proposition B.4. If R is entrywise nonnegative and irreducible then $B = (1-\kappa)I + \kappa R$ is entrywise nonnegative and irreducible, so by Perron-Frobenius its leading eigenvector is strictly positive and simple, and the binding mode is a genuine common mode in which every firm deviates in the same direction.

Entrywise nonnegativity of R is exactly the condition that no firm's feedback anti-aligns with another's. Shared vendors and shared pretraining corpora produce positive alignment, so the realistic regime satisfies it, and the paper states the condition rather than assuming it silently. This is (A5), and Proposition B.4 is the only place it is used.

B.5 Three anchors, and a mode swap

Monoculture. $R = \mathbf{1}\mathbf{1}^\top$ has rank one with eigenvalue N on the all-ones direction, so $\lambda_{\max} = N$ and $N_{\text{eff}} = 1 + \kappa(N-1)$. This is the inherited law whose predicted $2\times$ and $3\times$ at $N = 2, 3$ were measured at $1.74\times$ and $3.16\times$.

Orthogonal. $R = I$ gives $\lambda_{\max} = 1$, $N_{\text{eff}} = 1$ and $m_N = m_1$. A hundred firms perturbing the environment in a hundred orthogonal directions are dynamically one firm.

Simplex. With $r_{ij} = -1/(N-1)$ off the diagonal,

$$R_{\text{simplex}} \ = \ \tfrac{N}{N-1} I \ - \ \tfrac{1}{N-1}\mathbf{1}\mathbf{1}^\top,$$

whose diagonal is $N/(N-1) - 1/(N-1) = 1$ as required. Since $\mathbf{1}\mathbf{1}^\top$ has eigenvalue N on the all-ones direction and 0 on its complement, the spectrum is 0 on $\mathbf{1}$ and $N/(N-1)$ with multiplicity $N-1$. So $\lambda_{\max} = N/(N-1)$, $N_{\text{eff}} = 1 + \kappa/(N-1)$, and the radius tends to m_1 as N grows. The value $m_1(1-\kappa)$ does sit in the spectrum, carried by the all-ones direction at $\lambda = 0$, and it matches the inherited differential-mode eigenvalue derived by a different route, but it is not the spectral radius.

Under the simplex the all-ones direction carries the *smallest* eigenvalue, so the leading eigenvector is orthogonal to it and (A5) fails. That is consistent with Proposition B.4, since the simplex has negative entries and Perron-Frobenius does not apply. Certificate P4 checks all three parts, that the all-ones direction is annihilated, that the leading eigenvector is orthogonal to it, and that the entries are negative, at $N \in \{3, 5, 10, 30, 50\}$.

B.6 A witness for strict refinement

Proof of Proposition 3. Both markets satisfy (H1) to (H4) with the same response magnitude ε, the same objective curvature γ and sensitivity β, and the same spillover κ. They therefore share $m_1 = \varepsilon\beta/\gamma$ and share the joint map (1) in everything except R, so any quantity computed from response magnitudes, objective curvature and spillover alone takes the same value in both. By Theorem 2 the radii differ: $\lambda_{\max}(I) = 1$ gives $m_N = m_1$, while $\lambda_{\max}(\mathbf{1}\mathbf{1}^\top) = N$ gives $m_N = m_1(1+\kappa(N-1))$. The hypothesis $1/(1+\kappa(N-1)) < m_1 < 1$ puts the first below one and the second above it. The two markets thus agree on every scalar of the stated kind and disagree in stability, so a criterion that is a function of those scalars alone returns one verdict for both and is wrong about at least one. The interval is nonempty for every $N \geq 2$ and $\kappa > 0$. □

The witness is not a knife-edge construction. Any m_1 in an interval of length $1 - 1/N_{\text{eff}}$ works, which at $N = 20$ and $\kappa = 0.8$ is 94% of the unit interval.

B.7 Mean-alignment indices err, and always in the same direction

Proposition B.5. Let $\bar{m}$ be the off-diagonal mean of R. Then $\lambda_{\max}(R) \geq 1 + (N-1)\bar{m}$, hence $N_{\text{eff}} \geq 1 + \kappa(N-1)\bar{m}$, with equality if and only if $\mathbf{1}$ is a leading eigenvector of R, that is, if and only if R has constant row sums.

Proof. $\mathbf{1}^\top R\mathbf{1} = N + N(N-1)\bar{m}$, since the diagonal contributes N. The Rayleigh quotient at the all-ones vector gives $\lambda_{\max}(R) \geq \mathbf{1}^\top R\mathbf{1}/N = 1 + (N-1)\bar{m}$, with equality if and only if $\mathbf{1}$ attains the Rayleigh maximum, that is if and only if $R\mathbf{1}$ is parallel to $\mathbf{1}$. Multiplying by κ and adding $1-\kappa$ preserves the inequality. □

Certificate P5 finds zero violations on 360 random R and confirms equality on uniform R. Two readings follow. First, a mean-similarity diversity index never over-states systemic risk: it under-states it or is exact, so every error it makes is in the unsafe direction. A regulator using mean similarity is not merely imprecise, it is imprecise in the direction that lets unstable markets pass. Second, $\mathbf{1}^\top R\mathbf{1} = \|\sum_i \operatorname{vec}\hat{E}_i\|^2 \geq 0$ forces $\bar{m} \geq -1/(N-1)$, attained exactly at the simplex.

The gap is largest at the realistic topology, which is the vendor with a plurality rather than a monopoly. Take $N = 10$ with three firms fully aligned and every other pair orthogonal, so that R is block diagonal with a 3×3 all-ones block and $\lambda_{\max} = 3$. Three of the forty-five pairs are aligned, so $\bar{m} = 1/15$ and the mean index reports $\lambda_{\max} = 1.6$. At $\kappa = 0.8$ the true N_{eff} is 2.60 against the index's 1.48, an understatement by a factor of 1.757, and a market at $m_1 = 0.5$ then runs $m_N = 1.30$ and is unstable while the index reports 0.74 and calls it safe with margin.

The failure is not only one of level. The simplex *minimizes* the mean at $\bar{m} = -1/(N-1)$ while carrying $\lambda_{\max} = N/(N-1)$, which exceeds one, so it is spectrally worse than orthogonality even though it has the smaller mean. The mean therefore does not order configurations correctly, which is why every stability claim in this paper uses the spectral form and never a mean.

B.8 Heterogeneous moduli

Proposition B.6. Let $M = \operatorname{diag}(m_i)$ with every $m_i > 0$, and let $J = -M[(1-\kappa)I + \kappa R]$. Then

$$\max_i m_i \quad\leq\quad \rho(J) \quad\leq\quad \big(\max_i m_i\big)\, N_{\text{eff}}.$$

Proof. J is not symmetric, but the congruence $S = M^{1/2}$ gives $M^{1/2}(MB)M^{-1/2} = M^{1/2}BM^{1/2}$, so $\rho(J) = \lambda_{\max}(A)$ with $A = (1-\kappa)M + \kappa M^{1/2}RM^{1/2}$, symmetric and positive semidefinite. For the lower bound, $e_i^\top Ae_i = (1-\kappa)m_i + \kappa m_i r_{ii} = m_i$ since $r_{ii} = 1$, and the Rayleigh quotient at each e_i gives $\lambda_{\max}(A) \geq m_i$ for every i. For the upper bound, $R \preceq \lambda_{\max}(R)I$ in the Loewner order and congruence by $M^{1/2}$ preserves it, so $M^{1/2}RM^{1/2} \preceq \lambda_{\max}(R)M$ and hence $A \preceq N_{\text{eff}}M$, giving $\lambda_{\max}(A) \leq N_{\text{eff}} \max_i m_i$. □

The bound is exact at $\kappa = 0$ and at $R = I$, where $A = M$ and both sides equal $\max_i m_i$, and its upper half is exact at equal moduli, where it is Theorem 2. A mean-modulus formula is provably wrong: at $R = I$ the firms decouple and $\rho(J) = \max_i m_i$, certified with m ranging over $\{0.1, \ldots, 0.9\}$ at $\rho = 0.900$ against a mean of 0.400. Certificate P8 finds no violation on 300 random draws.

C Concentration to the supply-chain limit

The decomposition $E_i = \sqrt{s}\,E_{\text{shared}} + \sqrt{1-s}\,\Xi_i$ is meant to deliver $R \to (1-s)I + s\mathbf{1}\mathbf{1}^\top$ and therefore $N_{\text{eff}} \to 1 + \kappa s(N-1)$. Saying the cross terms vanish in expectation is not enough for a stability claim, because a market is unstable or not at the realized alignment matrix rather than at its mean.

(D) Design assumption. E_{shared} and the Ξ_i have independent entries, mean zero, unit variance, sub-Gaussian with a common constant, and the Ξ_i are independent of each other and of E_{shared}.

This is an idealization. Real response Jacobians are neither Gaussian nor independent across firms beyond the shared component, and what the argument uses is sub-Gaussian entries and independence of the idiosyncratic parts. Write $D = d^2$, $u = \operatorname{vec} E_{\text{shared}}$ and $\xi_i = \operatorname{vec} \Xi_i$, so that $\operatorname{vec} E_i = \sqrt{s}\, u + \sqrt{1-s}\, \xi_i$ in $\mathbb{R}^D$. Under (D), $\|u\|^2/D$ and $\|\xi_i\|^2/D$ concentrate at 1 while $u^\top \xi_j/D$ and $\xi_i^\top \xi_j/D$ concentrate at 0, all at rate $1/\sqrt{D}$, so $r_{ij} \to s$ for $i \neq j$ and $\lambda_{\max}(R_\infty) = 1 + s(N-1)$ exactly by the rank-one structure.

Lemma C.1. Under (D) there is an absolute constant C such that for any $\delta \in (0,1)$, with probability at least $1-\delta$,

$$\max_{i \neq j} \left| r_{ij} - s \right| \;\leq\; C \, \frac{\sqrt{\log(N/\delta)}}{d}.$$

Proof. Each of $\|u\|^2/D - 1$, $\|\xi_i\|^2/D - 1$, $u^\top \xi_i/D$ and $\xi_i^\top \xi_j/D$ is an average of D independent mean-zero sub-exponential terms, so Bernstein's inequality gives a tail $2\exp(-cDt^2)$ for $t \in (0,1)$. There are $O(N^2)$ such quantities across all pairs, and a union bound with $t = C\sqrt{\log(N/\delta)/D}$ controls all of them simultaneously with probability at least $1-\delta$. The ratio defining r_{ij} is a smooth function of these quantities with bounded derivatives in a neighbourhood of the limit point, so the same rate carries to $r_{ij} - s$, and $\sqrt{D} = d$ gives the statement. $\square$

The inner products average over d^2 matrix entries, so the fluctuation is $1/d$. Certificate P7 measures a log-log slope of -1.018 against a target of -1, at $N = 8$, $s = 0.6$, $d \in \{8, 16, 32, 64, 128\}$ and 40 trials per point.

Proposition C.2. Under the event of Lemma C.1,

$$\left| \lambda_{\max}(R) - (1 + s(N-1)) \right| \;\leq\; C \, \frac{N\sqrt{\log(N/\delta)}}{d},$$

$$\left| N_{\text{eff}} - (1 + \kappa s(N-1)) \right| \;\leq\; \kappa C \, \frac{N\sqrt{\log(N/\delta)}}{d}.$$

Proof. Write $\Delta = R - R_\infty$, which has zero diagonal since both matrices have unit diagonal. Weyl's inequality gives $|\lambda_{\max}(R) - \lambda_{\max}(R_\infty)| \leq \|\Delta\|_2$, and $\|\Delta\|_2 \leq \|\Delta\|_F \leq \sqrt{N(N-1)}\max_{i\neq j}|\Delta_{ij}| \leq N \max_{i \neq j} |r_{ij} - s|$. Apply Lemma C.1; the N_{eff} form follows on multiplying by κ. $\square$

The bound matters in relative terms, because N_{eff} itself grows with N. For large N the limit is $1 + s(N-1) \approx sN$, so the relative error is of order $C\sqrt{\log N}/(sd)$, with the N in the numerator cancelling against the N in N_{eff}. The closed form is therefore accurate once $d \gg \sqrt{\log N}/s$, a condition on dimension against vendor concentration rather than against firm count, and it degrades as $s \to 0$, which is correct: at low shared-model fraction the limit matrix is close to the identity and the leading eigenvalue is set by fluctuations rather than by the shared component. In the instantiation the base model calibrates, d is in the low hundreds and N in the tens, so the requirement is met with room for s above roughly 0.2.

The step $\|\Delta\|_2 \leq \|\Delta\|_F$ is crude and is used because it is elementary and unconditional. For a Δ with the Wishart-type structure that (D) produces, the operator norm concentrates at $\sqrt{N}/d$ rather than at N/d, which would improve the relative error to $1/(sd\sqrt{N})$, better for larger markets rather than worse. The sharp rate is reported as measured and not as proved, since claiming it otherwise would violate the paper's rule on status flags.

D The cadence composition

Lemma D.1. Let firm i retrain by gradient descent on a γ-strongly-convex quadratic own objective whose minimizer is the frozen best response b, with step size η. Then K steps from z_0 land at $z_K = b + c^K(z_0 - b)$ with $c = 1 - \eta\gamma$.

Proof. The gradient is $\gamma(z-b)$, so the update is $z \mapsto z - \eta\gamma(z-b) = b + (1-\eta\gamma)(z-b)$. Induct on K. $\square$

Here $c \in (0,1)$ requires $0 < \eta < 2/\gamma$, and $\eta \leq 1/\gamma$ gives $c \in [0,1)$. Certificate Q1 runs actual gradient descent and measures the realized gap ratio, exact to 10^{-12} at three (η, γ) pairs and $K \in \{1,3,7,20\}$. The constant c is never written into the loop; it is whatever the loop does.

Lemma D.2. For a single firm whose full-retraining best response has slope $-m$, the K-step outer map has slope $\mu(K) = -m + c^K(1+m)$.

Proof. The frozen best response at the current deployment h_t is $b = -mh_t$, so by Lemma D.1, $h_{t+1} = -mh_t + c^K(h_t + mh_t) = h_t[-m + c^K(1+m)]$. $\square$

At $K \to \infty$ this gives $-m$, the full-retraining slope, and at $K = 0$ it gives the identity, since a firm taking no gradient steps does not move. The inherited single-firm lemma is therefore reproduced rather than assumed.

The composition rests on one hypothesis, stated so that it can be checked.

(C) The contraction c is the same scalar in the joint market as in the single-firm market. It is built from own-objective curvature γ and inner step size η, neither of which involves N, κ or R: competitors change *where* firm i is heading, through the frozen best response, and not *how fast* it gets there.

This is the only place the composition can fail, so it is certified rather than argued. Certificate C12 measures the realized one-step contraction inside the joint market across 45 configurations spanning $N \in \{1,2,5,12,30\}$, $\kappa \in \{0, 0.5, 1\}$ and $m_1 \in \{0.05, 0.15, 0.6\}$ with random R at each, and the spread across all of them is 4.4×10^{-16}.

Proof of Theorem 4. All firms compute their frozen best responses from the common current state x, giving $b = Jx$, then each takes K inner steps. By Lemma D.1 applied coordinatewise, and using (C) so that the same c applies in every coordinate,

$$x_{t+1} \;=\; Jx_t + c^K(x_t - Jx_t) \;=\; \big[c^K I + (1-c^K)J\big]x_t.$$

This matrix is an affine function of J, so it shares J's eigenvectors, with eigenvalues $\mu_i(K) = c^K - (1-c^K)m_1\nu_i$ where $\nu_i = (1-\kappa) + \kappa\lambda_i(R)$. On the mode carrying $\lambda_{\max}(R)$, where $m_1\nu_{\max} = m_N$ by Theorem 2, this is $\mu_N(K) = c^K - (1-c^K)m_N = -m_N + c^K(1+m_N)$.

For the stability claim, $\mu_i(K)$ is strictly decreasing in ν_i because $(1-c^K)m_1 > 0$ for every $K \geq 1$. Every $\nu_i \geq 0$ by Lemma B.1(1) with $\kappa \leq 1$, so $\mu_i(K) \leq c^K < 1$ for all i and the upper side never binds, whatever R is; and $\min_i \mu_i(K)$ is attained at $\nu_{\max}$, so the constraint is exactly $\mu_N(K) > -1$. Expanding,

$$c^K - (1-c^K)m_N > -1 \iff m_N < \frac{1+c^K}{1-c^K} \iff c^K > \frac{m_N - 1}{m_N + 1},$$

and taking logarithms, with $\ln c < 0$ flipping the inequality on division, gives $K < K_{\max}$. When $m_N \leq 1$ the right-hand side of the last inequality is non-positive while $c^K > 0$, so the constraint is vacuous and every cadence is stable. $\square$

Certificate Q2 builds the joint map by running the actual inner gradient descent on each basis vector and comparing against $c^K I + (1-c^K)J$ at 32 combinations of (N, κ, m_1, K), agreeing below 10^{-12}; C8 checks the frontier on 51600 integer cases with zero mismatches; and Q5 iterates the actual joint map to convergence or divergence on 59 decisive trials at random (N, κ, s, m_1, K), with zero disagreements with the predicted side.

The monotonicity argument replaces an earlier route through the size of the differential modes. It is shorter and it is unconditional: it needs no Perron-Frobenius condition, because monotonicity places the extreme at $\lambda_{\max}$ regardless of the sign pattern of R.

Corollary D.3. $K_{\max}$ is strictly decreasing in m_N, and hence in s.

Proof. With $f(m) = \ln \frac{m-1}{m+1}$ we have $f'(m) = \frac{1}{m-1} - \frac{1}{m+1} = \frac{2}{m^2-1} > 0$ for $m > 1$, and dividing by $\ln c < 0$ flips the sign. Since $m_N = m_1(1 + \kappa s(N-1))$ is increasing in s, the composition is decreasing in s. ☐

Two readings the body uses. Since K is a positive integer, the realized window is $\lfloor K_{\max} \rfloor$ while the plotted frontier is continuous, and certificate Q4 confirms that $\lfloor K_{\max} \rfloor$ is the largest stable integer and $\lfloor K_{\max} \rfloor + 1$ is unstable at five (c, m_N) pairs. And a feasible cadence exists if and only if $K_{\max} > 1$, which is exactly $m_N < (1+c)/(1-c)$: past that level of effective crowding the market is unstable at every retraining frequency, and minimum-cadence operation multiplies the sustainable crowding by exactly $(1+c)/(1-c)$, a factor of 9 at $c = 0.8$. Critical crowding is therefore not a separate result but the $K = 1$ instance of the frontier.

Worked at $m_1 = 0.15$, $\kappa = 0.8$, $N = 30$ and $c = 0.8$, so that $N_{\text{eff}} = 1 + 23.2s$:

s	N_{eff}	m_N	$K_{\max}$
0.25	6.80	1.020	20.68
0.50	12.60	1.890	5.28
1.00	24.20	3.630	2.53

All three m_N values sit below the critical crowding level of 9, which is why every column has a finite window. The table is not reproducible without c, so $c = 0.8$ is stated wherever it appears. The operational reading is that the externality is not only that a competitor's entry consumes a firm's retraining budget, but that a competitor's choice of vendor does too, without their entering at all and without either firm doing anything wrong.

Two scope notes close this appendix. By Lemma D.1 the deployed model's gap to the frozen best response after a round is exactly c^K times the gap before it, so c^K is simultaneously what buys stability and what measures the staleness of the deployed model: the lever is not free, and its price is quantified in the same parameter that delivers its benefit. And at $m_N = 1$ exactly, $\mu_N(K) = -1 + 2c^K$ lies strictly inside $(-1, 1)$ for every finite K, so lazy retraining strictly stabilizes the marginally unstable market, with a margin $2c^K$ that decays geometrically and falls below double precision past $K = 171$ at $c = 0.8$. That last is a floating-point limit and not a statement about markets, and the certificate records it as such.

E The mixed market

Of N firms sharing one pool, N_b are blind and run the ordinary retraining loop while $N_{\text{corr}} = N - N_b$ are corrected and run the feedback-aware update governed by γ_{PO}. Write $\theta = \gamma/\gamma_{PO} \in (0, 1]$, so that $\theta = 1$ is no correction and $\theta \to 0$ is the strong-correction limit.

(M) Correction scales a firm's modulus: a corrected firm carries θm_1 where a blind firm carries m_1. This is the faithful reading of the single-firm result, in which the corrected loop's modulus is $m_1 \gamma/\gamma_{PO}$. Correction changes a firm's gain and not its response direction, so R is untouched.

Take the supply-chain alignment $R = (1-s)I + s\mathbf{1}\mathbf{1}^\top$ and write $k_s = \kappa s$, so that $B = (1 - k_s)I + k_s \mathbf{1}\mathbf{1}^\top$. By Proposition B.6 the radius of $J = -MB$ is $\lambda_{\max}(A)$ for $A = M^{1/2} B M^{1/2}$, and substituting B,

$$A \;=\; (1-k_s)M \;+\; k_s\, ww^\top, \qquad w = M^{1/2}\mathbf{1}, \quad w_i = \sqrt{m_i}.$$

So A is a diagonal matrix plus a positive-weight rank-one term, and its eigenvalues not equal to a diagonal entry solve the secular equation $1 + k_s \sum_i m_i/((1-k_s)m_i - \lambda) = 0$. With only two distinct moduli the sum collapses to two terms and the equation clears to a quadratic, which is why the mixed market has a closed form at all and why three or more correction levels would not.

Proof of Theorem 5. Write $a = (1-k_s)m_1$ and $b = (1-k_s)\theta m_1$ for the two diagonal values. Clearing denominators in the secular equation gives

$$(a-\lambda)(b-\lambda) + k_s N_b m_1 (b-\lambda) + k_s N_{\text{corr}} \theta m_1 (a - \lambda) = 0,$$

that is $\lambda^2 - P\lambda + Q = 0$ with

$$P = a + b + k_s m_1\big(N_b + \theta N_{\text{corr}}\big), \qquad Q = \theta m_1^2(1-k_s)\big(1+k_s(N-1)\big) = \theta m_1^2(1-k_s)N_{\text{eff}}.$$

The two secular roots are this quadratic's roots, and the eigenvectors constant within each block and orthogonal to w supply the rest of the spectrum, namely a with multiplicity $N_b - 1$ and b with multiplicity $N_{\text{corr}} - 1$. Because the rank-one weight k_s is positive, the eigenvalues of A interlace those of $(1-k_s)M$, so the largest eigenvalue of A exceeds every diagonal entry and is therefore the larger secular root rather than a degenerate one, giving $\rho(J) = \frac{1}{2}(P + \sqrt{P^2 - 4Q})$. □

Certificate H1 reports a maximum error of 2.5×10^{-14} against dense eigensolves over 6000 random draws. The appearance of N_{eff} inside Q is not cosmetic: the mixed market inherits Theorem 2's crowding through that product, which is why every statement here carries s.

Degenerate blocks. When $N_b = 0$ or $N_b = N$ one block is empty, but the quadratic still carries that block's factor and leaves a phantom root behind. At $N_b = 0$ the naive quadratic returns $(1-k_s)m_1$, a value no firm's modulus supports, and at small θ that phantom exceeds the true radius: 0.132 against a true 0.043 at $N = 12$, $m_1 = 0.3$, $\kappa = 0.8$, $s = 0.7$ and $\theta = 0.02$. The correct values are the single-block ones, $\rho = m_1 N_{\text{eff}}$ at $N_b = N$ and $\rho = \theta m_1 N_{\text{eff}}$ at $N_b = 0$, and any implementation must branch on this. Certified in H2.

Both limits are corollaries. At $\theta = 1$, $P = m_1[2(1-k_s) + k_s N]$ and $Q = m_1^2(1-k_s)N_{\text{eff}}$, whose roots are $m_1 N_{\text{eff}}$ and $m_1(1-k_s)$, recovering Theorem 2 exactly. As $\theta \to 0$, $Q \to 0$, so one root goes to zero and the other to $P \to m_1(1 + \kappa s(N_b - 1))$, the blind-block condition. The argument that corrected firms transmit no feedback and drop out of the cycle is right, and it is now a corollary of the exact root rather than a separate claim.

E.1 The limit errs, and it errs optimistic

Proposition E.1. $\rho(J)$ is nondecreasing in θ.

Proof. $A_{ij} = \sqrt{m_i m_j} B_{ij}$ and B is entrywise nonnegative for $k_s \in [0,1]$, so A is entrywise nonnegative. Every m_i is nondecreasing in θ, constant for blind firms and θm_1 for corrected ones, so every entry of A is nondecreasing in θ. Since A is symmetric and nonnegative, by Perron-Frobenius its Rayleigh maximizer may be taken in the nonnegative orthant, and for such an x the form $x^\top A(\theta)x$ is nondecreasing in θ. Taking x to be the maximizer at $\theta_1 < \theta_2$,

$$\lambda_{\max}(A(\theta_2)) \geq x^\top A(\theta_2)x \geq x^\top A(\theta_1)x = \lambda_{\max}(A(\theta_1)).$$ □

Two readings follow and they point opposite ways. The reassuring one is that correction never backfires: more correction is always weakly stabilizing, so there is no perverse configuration in which un-blinding a firm destabilizes the market. The uncomfortable one is that $\rho(\theta) \geq \rho(0)$, so the strong-correction limit *under-states* the radius. The limit is optimistic rather than conservative, and it can call a market stable that is unstable at any finite correction strength. On the random ensemble of Appendix K the limit calls 11.8% of configurations stable that are not, which is 2157 of 18313 draws with an exact 95% interval of $[11.3, 12.3]$%. The fraction depends on where the draws are taken from and the appendix states the ranges; the *direction* does not, since it follows from the monotonicity just proved and is observed on every seed with zero exceptions. Three worked cases, all of which the limit reads as comfortably stable:

Configuration	limit reads	truly unstable unless
$N = 10$, $N_b = 6$, $m_1 = 0.15$, $\kappa = 0.8$, $s = 1$	0.750	$\gamma_{PO} > 1.9\gamma$
$N = 30$, $N_b = 8$, $m_1 = 0.10$, $\kappa = 0.9$, $s = 0.8$	0.604	$\gamma_{PO} > 3.9\gamma$
$N = 20$, $N_b = 10$, $m_1 = 0.12$, $\kappa = 0.8$, $s = 1$	0.984	$\gamma_{PO} > 58.6\gamma$

Certificate C18 finds zero violations of Proposition E.1 on 3000 random configurations, each swept over 50 values of θ, and reports the verdict flips above. This is why the paper states the exact two-block root as the theorem and the limit as a corollary carrying its own error direction: shipping the limit alone would state a stability criterion that errs in the unsafe direction without saying so.

E.2 Imperfect correction

Proof of the imperfect-correction law and of Corollary 6. At $\kappa = s = 1$ the quadratic degenerates, since $1 - k_s = 0$ kills a, b and Q, leaving $\rho(J) = P = m_1(N_b + \theta N_{\text{corr}})$. Writing the corrected fraction as $\rho_c = 1 - N_b/N$ and solving $\rho(J) < 1$ for it gives

$$\rho_c^*(\theta) \;=\; \frac{1 - 1/m_N}{1-\theta} \;=\; \frac{1 - 1/m_N}{e}, \qquad e = 1 - \theta = 1 - \gamma/\gamma_{PO},$$

which is the standard epidemiological coverage requirement for an imperfect vaccine, with the clean law $1 - 1/m_N$ as its perfect-efficacy corner. The required fraction exceeds one, so that no corrected fraction stabilizes the market, exactly when $\theta > 1/m_N$, that is when $\gamma_{PO}/\gamma \leq m_N$, which is Corollary 6. □

Certificate H5 reports agreement to 1.8×10^{-15} over 120 configurations, with the radius form checked against dense eigensolves, and H6 certifies the critical efficacy. At $m_N = 2.5$ the corrected update must deliver at least a $2.5\times$ curvature improvement or the market cannot be stabilized by un-blinding at all, whatever fraction is treated. This is the exact structural parallel of the critical crowding level of Appendix D: each lever has a regime past which it stops working, and stating both is what makes the substitution frontier an honest object rather than an extrapolation.

The correspondence is worth being precise about. It now transfers not only the threshold but the standard refinement of the threshold, which is what a structural analogy predicts and a decorative one does not. Away from $\kappa = s = 1$ the closed form is no longer exact and the exact threshold comes from setting the quadratic's larger root to one; the law remains accurate to well under one firm's granularity at the parameters this paper uses.

E.3 The threshold in firms

Proposition E.2. The strong-correction limit is stable if and only if $N_b < N_c(s) = 1 + (1/m_1 - 1)/(\kappa s)$.

Proof. Rearrange $m_1(1 + \kappa s(N_b - 1)) < 1$. □

Certificate C14 finds zero mismatches on 4000 draws for this form. The clamped restatement is *not* equivalent: writing $\rho_c = 1 - N_b/N$ and $\rho_c^* = \max(0, 1 - N_c(s)/N)$, the criterion $\rho_c > \rho_c^*$ fails on 134 of those same 4000 draws, every failure at the same corner, namely $N_b = N$ with $N_c(s) \geq N$, the all-blind market that is stable because it needs no correction at all and that a clamp at zero combined with a strict inequality excludes. The paper therefore states $N_b < N_c(s)$ as the theorem, which is primitive and exact, and presents the clamped ρ_c^* as the policy object, read as the fraction that must be corrected, with $\rho_c^* = 0$ meaning no requirement rather than a strict inequality to clear.

Since $\rho_c^* N$ is generally not an integer and the panels run at N in the tens, where one firm is several percentage points, the realized requirement is

$$\text{minimum corrected firms} \;=\; N - \lceil N_c(s) \rceil + 1,$$

clamped to $[0, N]$, because the largest stable blind count is $\lceil N_c \rceil - 1$. The natural guess $\lceil \rho_c^* N \rceil$ agrees on every random draw, since random draws never land on an exact integer N_c, but it is off by one exactly when N_c is an integer: at $N = 20$, $m_1 = 0.15$, $\kappa = 0.8$ and the s giving $N_c = 8$, the truth is 13 corrected firms and the guess says 12. Certificate H4 checks the correct form exactly on 3000 draws. Worked at $N = 20$, $m_1 = 0.15$ and $\kappa = 0.8$:

s	$N_c(s)$	ρ_c^*	minimum corrected firms
1.0	8.08	0.596	12
0.5	15.17	0.242	5
0.2	36.42	0	0

At $\kappa = s = 1$ we have $N_c = 1/m_1$ and $N_{\text{eff}} = N$, so $\rho_c^* = 1 - 1/(m_1 N) = 1 - 1/m_N$, the epidemiological threshold with the systemic modulus as the reproduction number: a market of ten firms at $m_N = 2.5$ needs 60% un-blinded. And ρ_c^* is increasing in s, strictly so wherever it is positive, which is what makes model diversity and corrected learning substitutes and what the (ρ, s) frontier plots. Certified in C15 and C16.

Panel 4 runs at $N = 20$, $m_1 = 0.15$, $\kappa = 0.8$ and $s = 1$, where the imperfect-correction law is not exact, and measures thresholds of 12, 14, 16 and 20 firms at efficacies 1.00, 0.90, 0.75 and 0.60 against a law predicting 12, 14, 16 and 20. That is four out of four at the integer granularity a market actually faces. It is a dry run in the linearized reference environment and is recorded as evidence rather than as proof; the exact statement away from $\kappa = s = 1$ remains the quadratic's root.

F The Pigouvian wedge

F.1 The welfare object

Drive the linearized joint map with noise. Under (A1) and (A5) the binding mode is the common one, whose deviation z_t from the joint equilibrium obeys $z_{t+1} = -m_N z_t + \xi_t$ with ξ_t independent, mean zero and variance σ^2. This is an AR(1) with coefficient $-m_N$. Stationarity requires $|m_N| < 1$, which is the stability condition the rest of the paper works with, and taking variances of a stationary solution gives $V = m_N^2 V + \sigma^2$, hence

$$V(m_N) \; = \; \frac{\sigma^2}{1 - m_N^2}.$$

The coefficient is $-m_N$ and not m_N because the retraining Jacobian is $J = -m_1[(1-\kappa)I + \kappa R]$ and the common mode inherits its sign. The stationary variance is blind to that sign, since the recursion enters squared, which is why the same expression serves both. The sign is not lost information: it reappears in the lag-1 autocorrelation, which is $-m_N$ and therefore negative, so a crowded market shows oscillatory rather than persistent common-mode error. That is the observable the supervision estimator reads, and it is why a positive-autocorrelation diagnostic would look for the wrong thing. Certificate C19 checks both facts by simulation rather than asserting them.

V is the paper's price of instability, and it is a modelling choice: stationary variance is the smooth proxy for divergence risk, finite everywhere strictly inside the stable region and divergent exactly at the boundary the rest of the paper is about. Nothing below depends on the functional form beyond V being positive, increasing and convex in m_N on $[0, 1)$, all three of which the closed form satisfies.

(W1) Aggressiveness. Each firm chooses $a_i > 0$, meaning cadence, learning rate or responsiveness: anything that scales its own modulus. Write $m_i = \mu(a_i)$ with μ differentiable, strictly increasing and common across firms. Aggressiveness moves a firm's gain and not its response direction, so R and κ are held fixed.

(W2) Benefit. Aggressiveness buys tracking benefit $B(a_i)$, strictly increasing and strictly concave. Only firm i's own benefit depends on a_i, so the entire interaction runs through m_N.

(W3) Exchangeable alignment, with a simple leading eigenvalue. R is invariant under a common permutation of firms, $\kappa > 0$, and $\lambda_{\max}(R)$ is simple. The supply-chain alignment at any $s > 0$ is exchangeable with a simple leading eigenvalue, as is the monoculture corner. The orthogonal corner $R = I$ is excluded: there the coupling matrix is the identity, $\lambda_{\max}$ has multiplicity N, and the perturbation argument below has no unique v to differentiate along. Simplicity is what buys the uniform leading eigenvector at the symmetric point, and it is used in Lemma 7's proof rather than assumed away.

(W4) Client exposure. The planner's variance weight satisfies $W > \sum_i w_i$, with all weights positive.

(W4) is stated as an assumption and is not microfounded. Every trade has a client on the other side who bears execution-quality variance while no dealer's objective contains it, so the planner's weight on common-mode variance strictly exceeds what the dealers together internalize. The results below turn on the sign of $W - \sum_i w_i$ and never on its size, so a microfoundation would fix a magnitude the theorem does not use.

F.2 The marginal crowding share

Proof of Lemma 7. By Proposition B.6, for $M = \mathrm{diag}(m_1, \dots, m_N)$ and $B = (1-\kappa)I + \kappa R$ the radius of $J = -MB$ is $\lambda_{\max}(A)$ for $A(M) = M^{1/2}BM^{1/2}$. Since $dM^{1/2}/dm_i = (1/(2\sqrt{m_i}))e_i e_i^\top$,

$$\frac{dA}{dm_i} \;=\; \frac{1}{2\sqrt{m_i}}\Big(e_i e_i^\top BM^{1/2} + M^{1/2}Be_i e_i^\top\Big).$$

For a simple eigenvalue with unit eigenvector v, first-order perturbation gives $d\lambda_{\max}/dm_i = v^\top(dA/dm_i)v$. The two terms are transposes of one another and A is symmetric, so they contribute equally, giving $d\lambda_{\max}/dm_i = (BM^{1/2}v)_i v_i/\sqrt{m_i}$. At the symmetric point $M = m_1 I$ we have $A = m_1 B$, so v is the leading eigenvector of B and $BM^{1/2}v = \sqrt{m_1}\lambda_{\max}(B)v = \sqrt{m_1}N_{\text{eff}}v$, whence $d\lambda_{\max}/dm_i = N_{\text{eff}}v_i^2$. Summing over i gives N_{eff}, because v is a unit vector. Under (W3) an exchangeable B commutes with every permutation matrix, so a simple leading eigenvector must be permutation invariant, giving $v_i^2 = 1/N$ and $dm_N/dm_i = N_{\text{eff}}/N$. □

Certificate C34 checks this by finite differences on random alignment matrices, confirming that the shares sum to N_{eff} and equal N_{eff}/N on exchangeable R. It is the load-bearing certificate of this section: everything else in the theorem is arithmetic on top of the marginal crowding share, and until C34 that share was the step where a plausible-looking $1/N$ could have been wrong by a factor of N_{eff}.

Two things are worth reading off. The *sum* of the marginal effects is N_{eff} and not 1, so the market's total sensitivity to a uniform increase in aggressiveness is amplified by exactly the effective learner count. And an *individual* firm moves m_N by N_{eff}/N, so as N grows each firm's own footprint shrinks while the total grows, which is the arithmetic shape of every commons problem.

F.3 The two problems and the wedge

Firm i bears weight w_i on common-mode variance and the planner bears W:

$$\text{private:}\ \max_{a_i} B(a_i) - w_i V(m_N(a)), \qquad \text{social:}\ \max_{a} \sum_j B(a_j) - WV(m_N(a)).$$

Write $V'(m) = 2\sigma^2 m/(1-m^2)^2 > 0$ on $(0,1)$ and evaluate at a symmetric profile $a_i = a$, so that $m_N = \mu(a)N_{\text{eff}}$. Chaining (W1) with Lemma 7 gives $dm_N/da_i = (N_{\text{eff}}/N)\mu'(a)$ at the symmetric profile, and the two first-order conditions are

$$\text{(P)}\quad B'(a) = w_i V'(m_N)\frac{N_{\text{eff}}}{N}\mu'(a), \qquad \text{(S)}\quad B'(a) = WV'(m_N)\frac{N_{\text{eff}}}{N}\mu'(a).$$

They are the same equation with a different price on crowding, and the theorem is the gap between the two coefficients.

Proof of Theorem 8. Subtract (P) from (S). A firm facing a per-unit fee t on aggressiveness solves $\max B(a_i) - w_i V(m_N(a)) - ta_i$, whose first-order condition is (P) with $B'(a)$ replaced by $B'(a) - t$. Substituting

$$t^* = (W - w_i)V'(m_N)\frac{N_{\text{eff}}}{N}\mu'(a) = (W - w_i)\frac{2\sigma^2 m_N}{(1-m_N^2)^2}\cdot\frac{N_{\text{eff}}}{N}\mu'(a)$$

reproduces (S) term for term, so the taxed decentralized equilibrium implements the symmetric social optimum. □

With symmetric weights $w_i = w$ and $W = \chi N w$ for the $\chi > 1$ that (W4) supplies, $t^* = (\chi N - 1)wV'(m_N)(N_{\text{eff}}/N)\mu'(a)$ against a total marginal cost of $\chi N w V'(m_N)(N_{\text{eff}}/N)\mu'(a)$, so the ignored fraction is $(\chi N - 1)/(\chi N)$. Setting $\chi = 1$, which shuts off client exposure entirely and leaves only the firm-to-firm channel, the ignored fraction is exactly $(N-1)/N$. Note that N_{eff} cancels from this ratio, so the $(N-1)/N$ reading is a generic commons result and it is the provenance asymmetry below that is specific to this paper.

Proposition F.1. t^* is strictly increasing in N, in κ and in m_N, and diverges as $m_N \to 1$.

Proof. Take the supply-chain specialization $N_{\text{eff}} = 1 + \kappa s(N-1)$ with symmetric weights, so that $t^* = (\chi - 1/N)wV'(m_N)N_{\text{eff}}\mu'(a)$. The factor $\chi - 1/N$ is strictly increasing in N; N_{eff} is strictly increasing in N whenever $\kappa s > 0$ and in κ whenever $s > 0$ and $N \geq 2$; $m_N = \mu(a)N_{\text{eff}}$ inherits both; and $V'(m) = 2\sigma^2 m/(1-m^2)^2$ is strictly increasing on $(0,1)$ and diverges at 1, since both m and $(1-m^2)^{-2}$ are. Every factor moves the same way, so the product does. ☐

The divergence is the economically loaded part. The fee that would correct a market is not proportional to how crowded it is; it blows up like $(1-m_N)^{-2}$ as the market approaches its stability boundary, so a regulator setting a fee from a comfortable-looking market and holding it fixed is setting it far too low by the time the market is close to the edge. Certified in C20.

Proof of Corollary 9. Write $C(a) = V(m_N(a))$ and

$$F_{\text{priv}}(a) = B'(a) - wC'(a), \qquad F_{\text{soc}}(a) = B'(a) - WC'(a).$$

Both are strictly decreasing, since B' is strictly decreasing by (W2) and C' is nondecreasing by convexity, so each has at most one zero and by the interior assumption exactly one, namely a_d and a_s. Because $C'(a) > 0$ on the stable range, as $V' > 0$, $N_{\text{eff}} > 0$ and $\mu' > 0$, and because $W > w$ by (W4),

$$F_{\text{soc}}(a) = F_{\text{priv}}(a) - (W-w)C'(a) < F_{\text{priv}}(a) \quad \text{for every } a.$$

Evaluating at a_d gives $F_{\text{soc}}(a_d) < F_{\text{priv}}(a_d) = 0$, and since F_{soc} is strictly decreasing and vanishes at a_s, this forces $a_s < a_d$. Setting $\chi = 1$ so that $W = Nw$ gives $W - w = (N-1)w > 0$ exactly when $N \geq 2$, so the strict inequality survives with no client-side channel at all. At $N = 1$ with $\chi = 1$ the wedge is zero and the two problems coincide, which is the correct degenerate case: a single firm bearing its own variance in full internalizes everything. ☐

The distortion grows as the market crowds, since the coefficient gap $(W-w)C'(a)$ is proportional to $V'(m_N)N_{\text{eff}}$, both of which increase in N and in κ. Certified in C21, including the vanishing gap at $N = 1$.

F.4 The provenance channel

Proposition F.2. Under the supply-chain specialization, $dm_N/ds = m_1\kappa(N-1)$ exactly, and the corresponding adoption wedge is $t_s^* = (W - w_i)V'(m_N)m_1\kappa(N-1)$.

Proof. $m_N = m_1(1 + \kappa s(N-1))$ is affine in s, so the derivative is the coefficient, and the wedge follows from the proof of Theorem 8 with s in place of a_i as the choice variable. ☐

The same wedge therefore prices shared-model adoption directly, which turns the monoculture externality into a fee rather than a warning. The contrast with the aggressiveness channel is the paper-specific result. The aggressiveness derivative carries a factor N_{eff}/N, bounded above by $(1+\kappa(N-1))/N$ and therefore tending to κ from above as N grows, so a firm's marginal footprint through its own cadence does not grow without bound. The provenance channel is linear in N and does not decay. A firm's decision to adopt the market-leading vendor imposes a marginal cost on the market that grows with market size while its private quality gain does not, and that asymmetry is why unregulated adoption dynamics run toward the unstable configuration rather than away from it, and why a diversity floor is a different instrument from a cadence cap rather than a restatement of it. Certificate W6 measures the provenance channel growing from 0.79 to 87.91 between $N = 10$ and $N = 1000$ while the aggressiveness channel falls from 0.82 toward $\kappa = 0.8$.

The three levers are the standard instrument taxonomy: cadence caps are quantity regulation, correction mandates are a technology mandate, diversity floors are a structural remedy, and the wedge is a price instrument. The choice between prices and quantities under uncertainty is Weitzman's question, which this paper cites for framing rather than solves.

G Supervision from public prices

Every quantity in Theorems 2 through 8 involves R and s, which no outsider observes. Supervisors do not inspect firms' models, training sets or vendor contracts, so a theory whose every variable is unobservable to the regulator it addresses is a theory of a problem rather than a theory a regulator can act on.

Near the boundary the system tells on itself. As $m_N \to 1$ the aligned combination of firms' decisions exhibits critical slowing down: the lag-1 autocorrelation of the leading principal component of public quotes approaches one in magnitude, negative because the common mode oscillates, and its variance share grows like $1/(1 - m_N)$. A supervisor can therefore estimate distance to instability from public prices alone, with no access to any firm's model, data or code, and the estimator sharpens exactly as the market approaches the boundary, which is the opposite of the usual situation in which risk measurement degrades in the regime that matters.

This material is deliberately thin and the body flags it as deferred. The body carries the estimator and the consistency claim in one paragraph. The empirical panel, namely the leading component's variance share and lag-1 autocorrelation of cross-sectional spread co-movement overlaid on the model-implied fragility index, together with a placebo on series where no dealer-model channel exists, is scoped and is not run for this submission. It would be framed as consistency evidence and never as identification, since spread co-movement is contaminated by common macro shocks and the data are public proxies rather than trade-level records. The consistency proof, the central limit theorem and the detection sample complexity are journal material. Writing the estimator down formally, rather than as the description above, is the first open item.

H Certificates

Every closed form in the body carries a numerical certificate. A certificate is an assertion-based script that re-derives the expectation independently, by dense eigensolve, finite difference, brute force or direct simulation, rather than importing the theory module's own answer, and that exits nonzero on any disagreement. A closed form reaching the paper without a passing certificate ships as derived and never as verified.

Assertions	File	What it establishes
123	`verify_theorem1_proof.py`	the reduction, the properties of R, the spectrum, the mean bound, the heterogeneous-modulus bound, and the supply-chain concentration rate
60	`verify_theorem1_anchors.py`	the three anchors, the clustered topology, and the signed error of the mean index
59	`verify_theorem2_cadence.py`	the inner contraction, hypothesis (C), the joint map, the frontier in both forms, and the realized dynamics
70	`verify_theorem3_herd_immunity.py`	the two-block quadratic, the degenerate blocks, monotonicity in θ, the imperfect-correction law, the critical efficacy and the integer threshold
125	`verify_theorem4_wedge.py`	the AR(1) reduction and its sign convention, Lemma 7 by finite differences, the comparative statics, and over-adaptation
56	`verify_theory_module.py`	acceptance tests on the closed-form module
32	`verify_hetero_env.py`	the reference environment, including its reduction to the homogeneous market at $R = \mathbf{1}\mathbf{1}^\top$
17	`verify_ensemble_intervals.py`	the sampling measure, the count and the exact interval behind the one random-ensemble fraction the paper quotes
542	total	all passing at submission

Two of these were falsifying rather than confirming. C18 showed that the strong-correction limit errs optimistic, which changed what Theorem 5 claims and moved the exact two-block root from an optional refinement to the theorem itself. C14 showed that the clamped restatement of the blind-firm threshold is not equivalent to the primitive form, failing on 134 of 4000 draws at the all-blind corner. Both corrections are recorded above, where they apply.

I Experimental specifications

Environments. There are two, and the distinction between them is what the status flags track. The *order-flow simulator* is a genuine N-dealer market with informed flow, spread and inventory, which reduces bit for bit to the single-dealer market at $N = 1$. The *reference environment* places firms at chosen response directions and lets them retrain against a shared pool, realizing the response geometry of Lemma 1 and nothing else: no informed flow, no spread, no inventory, no microstructure of any kind. Its acceptance test, run before any measurement is taken from it, is that at $R = \mathbf{1}\mathbf{1}^\top$ it reproduces the homogeneous market on three counts, namely a measured alignment of exactly $\mathbf{1}\mathbf{1}^\top$, an N_{eff} of exactly $1 + \kappa(N-1)$, and a built Jacobian equal to (1) to 10^{-10}.

Protocol. Inherited from the base model and non-negotiable, since violating any of these silently invalidates a panel: sweep the feedback gain and never the confounded adversariality parameter; probe at the operating spread with common random numbers; scale the liquidity boost per the environment's guidance when multi-dealer runs approach the informed-flow cap, and never de-saturate silently; and treat beyond-boundary probe readings as diagnostics rather than as slopes. All runs are CPU-only and deterministic from a (config, seed) pair. Stability is estimated from the trajectory's asymptotic growth rate under the actual retraining map rather than read off an eigensolve, so a panel can disagree with its closed form.

The panels. Panel 1 sweeps N in the homogeneous shared-pool market and reproduces the amplification reported in a recent preprint. It is the paper's one measured result, and what it validates is the monoculture corner that Theorem 2 generalizes rather than any of the four results built on it. Its differential mode measures 3.4×10^{-3} against a theoretical zero at $\kappa = 1$, so the instability it exhibits is purely common-mode, which is the mechanism Theorem 2 generalizes. Panel 2 measures m_N over a grid of firms by shared-model fraction against the predicted boundary, with firms placed at the target alignment exactly rather than in distribution so that an s sweep moves along the exact curve; its clustered companion, three aligned firms among ten, is the topology worked in Appendix B. Panel 3 sweeps the (N_{eff}, K) grid against $K_{\max}$ at $c = 0.8$, including the critical-crowding column where the window closes. Panel 4 runs mixed markets at corrected fractions $0, 1/N, \ldots, 1$ in an unstable market. Panel 5 traces the (ρ_c, s) iso-stability curve, and it is the paper's most policy-legible object. Panel 6 solves both first-order conditions by bisection with m_N read from the reference environment's realized dynamics rather than from the closed form, so the over-adaptation verdict is a property of the simulated market.

Why five of six are dry runs. A heterogeneous-response port of the order-flow simulator was built and is exact at flat exposure profiles, reproducing the unmodified simulator at every N from 1 to 6 at relative error 0.00. Its acceptance gate then failed. The liquidity and price-impact channels stay shared whatever the response profiles are, leaving a residual reaching 17.7 percentage points across the separation range at $N = 2$, larger than several effects these panels resolve, and at $N \geq 4$ the finite-difference probe no longer measures a local slope. Two repairs are identified, routing liquidity and impact through the profiles as well, or finding a configuration whose probe stays local at the N the panels need, and neither is attempted for this submission, since both are multi-day edits and the first risks the bit-for-bit anchor that panel 1 rests on. No figure is drawn from that port and no status was moved. Panel 6 is a dry run for a different reason: the order-flow simulator carries no aggressiveness choice variable and no welfare object, so a measured counterpart does not exist to be built.

J Deferred items

Stated so that the boundary of what is proved is legible. The fully heterogeneous case without (H1), which is the Kronecker object of Appendix B, and share-weighted alignment for unequal-sized firms, which needs a majorization condition on the share vector since the naive claim that any reweighting raises $\lambda_{\max}$ is false. Non-separable spillover κ_{ij}, which folds into R only when it factors. The sharp operator-norm rate $\sqrt{N}/d$ of Appendix C, which needs a matrix-Bernstein argument rather than the Frobenius step, and a version of design assumption (D) under dependence across the idiosyncratic components, which is what shared data rather than shared models would produce. Asynchronous and heterogeneous cadences, which break the eigenvector sharing that Theorem 4 uses. Three or more correction levels, which break the two-block collapse and return a genuine secular equation. A corrected firm that also changes its response direction, which would move R and which hypothesis (M) does not model. And beyond the symmetric point, the observation that $dm_N/dm_i = N_{\text{eff}} v_i^2$ makes the marginal crowding share the leading eigenvector's participation ratio at firm i, so a firm sitting on the crowded mode imposes more than its $1/N$ share, and the asymmetric wedge that follows is a genuine open question.

K Statistical protocols

The paper makes two kinds of empirical statement and they are held to different standards, stated here once and used everywhere.

Deterministic grid checks keep worst-case language. Panels 2 to 6 compare a closed form against realized dynamics over a stated grid of configurations, with the run deterministic from a (config, seed) pair. What is reported is the maximum departure over the whole grid, not a mean with an interval. That is the stronger statement of the two, since it bounds every cell rather than describing a typical one, and dressing it in confidence intervals would weaken it while looking more statistical. No interval appears anywhere in Table 1's dry-run rows for that reason.

Random-ensemble fractions carry a sampling measure, an n, an exact count and an exact interval. The paper quotes one such fraction: how often the strong-correction limit of Section 6 calls a configuration stable that is unstable at every finite correction strength. Its protocol, fixed before the number was read and unchanged since:

- *Sampling measure.* N uniform on the integers 2 to 39; the blind count N_b uniform on 1 to $N-1$, so both blocks are non-empty and the two-block quadratic is the right form; m_1 uniform on $[0.02, 0.9]$; κ and s each uniform on $[0.05, 1]$; and the correction parameter $\theta = \gamma/\gamma_{PO}$ uniform on $[0.01, 1]$, so the ensemble spans efficacies from 0 to 0.99.
- *Estimator.* For each draw, the spectral radius is computed twice by dense eigensolve, once at $\theta \to 0$ and once at the drawn θ. A draw counts as a flip when the first is below one and the second is not. The eigensolve is independent of the closed-form quadratic, so this is a second route to the radius rather than a restatement of it.
- *Interval.* Clopper-Pearson at 95%, from beta quantiles rather than a normal approximation, so no large-np condition is needed.
- *Reported.* 2157 flips of 18313 draws, 11.8%, interval $[11.3, 12.3]\%$. Five independent ensembles of 20000 draws each give 11.74, 11.76, 12.01, 11.46 and 11.53%, every one inside that interval.
- *Protocol dependence, stated rather than hidden.* The fraction is a property of the ensemble as much as of the theorem. Widening the m_1 range to $[0.02, 1.2]$ moves it to 9.6%. The ranges above are therefore part of the claim, and the number should be read as the flip rate on this ensemble rather than as a universal constant.

What is not statistical here. The *direction* of the limit's error is a theorem, not an estimate: ρ is nondecreasing in θ by Perron-Frobenius on an entrywise-nonnegative matrix (Appendix E), so the limit can only under-state the radius. Across the five ensembles above, zero draws erred the other way, which is what a proved direction looks like when it is sampled and is not offered as evidence for it. Certificate `verify_ensemble_intervals.py` carries all of the above as 17 assertions, including the protocol itself, so the interval in the body is a function of a certified count rather than of a fresh draw.

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Justification: The public repository names an author, so it is withheld at submission to preserve the blind. The derivations are in Appendices A–F and the certificate inventory in Appendix H, both in this PDF. The seven certificate files themselves and the panel code go to reviewers as anonymized supplementary material, and the repository is linked in the camera-ready version; Section 8 states this arrangement. No dataset is used: every panel is simulated and deterministic from a (config, seed) pair.

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Justification: All experiments are CPU-only and numpy-only. All seven certificate files and the full six-panel suite together run in about twenty seconds on a single commodity laptop core, with no GPU, no accelerator and no distributed compute. No unreported large-scale preliminary experiments were run.

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